MATHS MANTHAN

• TUITION CLASSES •

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Subject: Mathematics | **PERIODIC TEST – 04 (SOLUTIONS)** | **Time:** 60 Minutes

Class: Class X (CBSE) | **Chapter 12: Surface Areas and Volumes** | **MM:** 24

Document Type: Official Step-by-Step Marking Scheme | **Published for:** mathsmanthan.co.in

CBSE STEP-MARKING GUIDELINES & EVALUATION PROTOCOL:

1. Full marks require explicit formula declaration, factoring common terms like πr before substituting $\pi = \frac{22}{7}$ or 3.14, showing intermediate arithmetic clearly, and boxing the final answer with proper physical units ($\text{cm}^2, \text{cm}^3, \text{m}^2, \text{m}^3$).
2. Partial step marks are allocated in brackets (**[0.5 Mark], [1 Mark], [1.5 Marks], [2 Marks]**). Alternative valid methods receive full credit.
3. Total marks of printed questions Q1–Q11 $= 5 + 4 + 6 + 5 + 4 = 24$ Marks (matching section totals; source paper header prints MM: 30).

Official Evaluation Key | Maths Manthan Meerut

SECTION A: Objective & MCQs

MATHS MANTHAN | ($5 \times 1 = 5$ Marks)

Q1. The ratio of total surface area of a solid hemisphere to the square of its radius is: [1]

(A) $2\pi : 1$ (B) $3\pi : 1$ (C) $4\pi : 1$ (D) $1 : 3\pi$

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B) $3\pi : 1$** [0.5 Mark]

Formula: A solid hemisphere includes its curved area $2\pi r^2$ and flat circular base πr^2.

Formula / Theorem Used: TSA of Solid Hemisphere $= 3\pi r^2 \implies \frac{\text{TSA}}{r^2} = \frac{3\pi r^2}{r^2} = \boxed{3\pi : 1}$

Q2. If the volume of a sphere of radius R is 16 times the volume of a hemisphere of radius r, then the ratio $\frac{R}{r}$ is: [1]

(A) 1 (B) 2 (C) 3 (D) 4

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(B) 2** [0.5 Mark]

Given: $V_{\text{sphere}}(R) = 16 \times V_{\text{hemi}}(r)$.

Formula / Theorem Used: $\frac{4}{3}\pi R^3 = 16 \times \frac{2}{3}\pi r^3$

$$\frac{4}{3}R^3 = \frac{32}{3}r^3 \implies R^3 = 8r^3 \implies \left(\frac{R}{r}\right)^3 = 2^3 \implies \boxed{\frac{R}{r} = 2}$$

Q3. If the diagonal of a cube is 17.32 cm, its volume is (Take $\sqrt{3} = 1.732$): [1]

(A) 1000 cm^3 (B) 1732 cm^3 (C) 173.2 cm^3 (D) 10000 cm^3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: **(A) 1000 cm^3** [0.5 Mark]

Formula: Space diagonal of cube of side a is $a\sqrt{3}$.

Formula / Theorem Used: $a\sqrt{3} = 17.32 \implies a = \frac{17.32}{1.732} = 10 \text{ cm} \implies V = a^3 = 10^3 = \boxed{1000 \text{ cm}^3}$

Q4. The surface area of a cube is 216 cm^2. Its volume is: [1]

(A) 144 cm^3 (B) 196 cm^3 (C) 212 cm^3 (D) 216 cm^3

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: $\boxed{\textbf{(D) } 216 \textbf{ cm}^3}$ [0.5 Mark]

Formula: TSA of Cube $= 6a^2$ and Volume $= a^3$.

$$6a^2 = 216 \implies a^2 = \frac{216}{6} = 36 \implies a = 6 \text{ cm} \implies V = 6^3 = \boxed{\textbf{216 cm}^3}$$

Q5. Assertion (A): Two cubes each of volume 125 cm^3 are joined end-to-end to form a cuboid. The surface area of the resulting cuboid is 250 cm^2.
Reason (R): If two identical cubes of edge a are joined end-to-end, the total surface area of the resulting cuboid is $10a^2$. [1]

(A) Both (A) and (R) are true, and (R) is the correct explanation of (A).

(B) Both (A) and (R) are true, but (R) is not the correct explanation of (A).

(C) (A) is true, but (R) is false. (D) (A) is false, but (R) is true.

▶ **STEP-BY-STEP SOLUTION:** [1 Mark]

Correct Option: $\boxed{\textbf{(A)}}$ [0.5 Mark]

Verification: Joining 2 cubes gives $L = 2a, B = a, H = a \implies \text{TSA} = 2(2a^2 + a^2 + 2a^2) = 10a^2$ (Reason (R) is true).
Here $a^3 = 125 \implies a = 5$ cm, so TSA $= 10(5)^2 = 250$ cm^2. Both (A) and (R) are true, and (R) explains (A).

SECTION B: Very Short Answer Questions MATHS MANTHAN | (2 × 2 = 4 Marks)

Q6. If the volumes of two cones are in the ratio 1 : 4 and their diameters are in the ratio 4 : 5, write the ratio of their heights. [2]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Diameter to Radius Ratio [1 Mark]

Since $\frac{d_1}{d_2} = \frac{r_1}{r_2} = \frac{4}{5}$ and $\frac{V_1}{V_2} = \frac{1}{4}$:

Formula / Theorem Used: $\frac{V_1}{V_2} = \left(\frac{r_1}{r_2}\right)^2 \left(\frac{h_1}{h_2}\right) \implies \frac{h_1}{h_2} = \frac{V_1}{V_2}\left(\frac{r_2}{r_1}\right)^2$

Step 2: Calculation [1 Mark]

$$\frac{h_1}{h_2} = \frac{1}{4} \times \left(\frac{5}{4}\right)^2 = \frac{1}{4} \times \frac{25}{16} = \frac{25}{64} \implies \boxed{\textbf{25 : 64}}$$

Q7. A solid sphere and a solid cube have equal surface areas. What is the ratio of the volume of the sphere to that of the cube? [2]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 2 Marks]

Step 1: Equating Surface Areas [1 Mark]

Let sphere radius $= r$ and cube edge $= a$. Given $4\pi r^2 = 6a^2 \implies \frac{r}{a} = \sqrt{\frac{6}{4\pi}} = \sqrt{\frac{3}{2\pi}}$.

Formula / Theorem Used: $\frac{V_{\text{sphere}}}{V_{\text{cube}}} = \frac{\frac{4}{3}\pi r^3}{a^3} = \frac{4}{3}\pi\left(\frac{r}{a}\right)^3$

Step 2: Simplifying Volume Ratio [1 Mark]

$$\frac{V_{\text{sphere}}}{V_{\text{cube}}} = \frac{4}{3}\pi\left(\frac{3}{2\pi}\right)\sqrt{\frac{3}{2\pi}} = 2\sqrt{\frac{3}{2\pi}} = \sqrt{\frac{6}{\pi}}$$

Substituting $\pi = \frac{22}{7}$: $\sqrt{\frac{42}{22}} = \sqrt{\frac{21}{11}} \implies$ $\boxed{\sqrt{6} : \sqrt{\pi} \ \text{(or } \sqrt{21} : \sqrt{11} \approx 1.38 : 1)}$.

SECTION C: Short Answer Questions — MATHS MANTHAN | (2 × 3 = 6 Marks)

Q8. The interior of a building is in the form of a cylinder of diameter 4.3 m and height 3.8 m, surmounted by a cone whose vertical height is 2.1 m. Find the total interior surface area and volume of the building. [3]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Dimensions & Slant Height [1 Mark]

Radius $r = \frac{4.3}{2} = 2.15$ m, $h_{\text{cyl}} = 3.8$ m, $h_{\text{cone}} = 2.1$ m.

$l = \sqrt{r^2 + h_{\text{cone}}^2} = \sqrt{(2.15)^2 + (2.1)^2} = \sqrt{4.6225 + 4.41} = \sqrt{9.0325} \approx 3.005$ m.

Formula / Theorem Used: Interior SA (walls + roof) $= \pi r(2h_{\text{cyl}} + l)$, $\quad V = \pi r^2\left(h_{\text{cyl}} + \frac{h_{\text{cone}}}{3}\right)$

Step 2: Interior Surface Area & Volume ($\pi = \frac{22}{7}$) [2 Marks]

$$\text{SA} = \frac{22}{7} \times 2.15 \times (7.6 + 3.0054) = \frac{22}{7} \times 2.15 \times 10.6054 \approx \boxed{71.66 \text{ m}^2} \ \text{(plus floor } \pi r^2 : 86.19 \text{ m}^2)$$

$$V = \frac{22}{7} \times (2.15)^2 \times \left(3.8 + \frac{2.1}{3}\right) = \frac{22}{7} \times 4.6225 \times 4.5 = \boxed{65.38 \text{ m}^3}$$

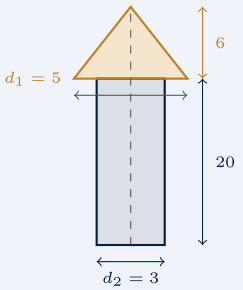
Q9. A wooden toy rocket is in the shape of a cone mounted on a cylinder. The height of the entire rocket is 26 cm, while the height of the conical part is 6 cm. Base diameter of cone is 5 cm, while base diameter of cylinder is 3 cm. If the cone is painted orange and cylinder yellow, find the area painted with each colour. (Use $\pi = 3.14$) [3]

► **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 3 Marks]

Step 1: Dimensions & Slant Height [1 Mark]

Cone: $r_1 = 2.5$ cm, $h_1 = 6$ cm $\implies l = \sqrt{2.5^2 + 6^2} = \sqrt{6.25 + 36} = \sqrt{42.25} = 6.5$ cm.

Cylinder: $r_2 = 1.5$ cm, height $h_2 = 26 - 6 = 20$ cm.



Formula / Theorem Used: Orange $= \pi r_1 l + \pi(r_1^2 - r_2^2)$, $\quad$ Yellow $= 2\pi r_2 h_2 + \pi r_2^2$

Step 2: Painted Areas ($\pi = 3.14$) [2 Marks]

Orange $= 3.14 \times (2.5 \times 6.5 + 2.5^2 - 1.5^2) = 3.14 \times (16.25 + 6.25 - 2.25) = 3.14 \times 20.25 = \boxed{63.585 \text{ cm}^2}$.

Yellow $= 3.14 \times (2 \times 1.5 \times 20 + 1.5^2) = 3.14 \times (60 + 2.25) = 3.14 \times 62.25 = \boxed{195.465 \text{ cm}^2}$.

Q10. A vessel is in the form of an inverted cone. Its height is 8 cm and the radius of its top, which is open, is 5 cm. It is filled with water up to the brim. When lead shots, each of which is a sphere of radius 0.5 cm, are dropped into the vessel, one-fourth of the water flows out. Find the number of lead shots dropped into the vessel. [5]

▶ **STEP-BY-STEP SOLUTION & MARKING SCHEME:** [Total: 5 Marks]

Step 1: Dimensions & Overflow Volume Formula [1.5 Marks]

Cone: $R = 5$ cm, $H = 8$ cm. Lead shot: $r = 0.5$ cm $= \frac{1}{2}$ cm. Water spilled $= \frac{1}{4}V_{\text{cone}}$.

Formula / Theorem Used: $n \times \frac{4}{3}\pi r^3 = \frac{1}{4}\left(\frac{1}{3}\pi R^2 H\right)$

Step 2: Factoring π & Computing Volumes [2 Marks]

$V_{\text{spilled}} = \frac{1}{4} \times \frac{1}{3}\pi(5)^2(8) = \frac{1}{12}\pi \times 25 \times 8 = \frac{50\pi}{3}$ cm^3.

$V_{1\text{ shot}} = \frac{4}{3}\pi\left(\frac{1}{2}\right)^3 = \frac{4}{3}\pi \times \frac{1}{8} = \frac{\pi}{6}$ cm^3.

Step 3: Number of Lead Shots [1.5 Marks]

$$n = \frac{V_{\text{spilled}}}{V_{1\text{ shot}}} = \frac{\frac{50\pi}{3}}{\frac{\pi}{6}} = \frac{50}{3} \times \frac{6}{1} = \boxed{\textbf{100 lead shots}}$$

Q11. Case Study: Adventure Camp Conical Tents and Refreshments

Students at an adventure camp in Sakleshpur were served juice upon arrival: some in a cylindrical glass ($r = 3.5$ cm, $h = 10.5$ cm) and some in a hemispherical cup ($r = 3.5$ cm). Later, each group of four students was given 551 m^2 canvas to erect a conical tent of base radius 7 m, with 1 m^2 wasted during stitching. [4]

(i) Find the capacity of the cylindrical glass. [1]

(ii) Find the capacity of the hemispherical cup. [1]

(iii) Find the height of the conical tent constructed with the available canvas. [1]

(iv) How much floor area is occupied by each student inside the tent? [1]

▶ **CASE-BASED SOLUTION & MARKING SCHEME:** [Total: 4 Marks]

Extracted Scenario Data: Glass: $r = 3.5 = \frac{7}{2}$ cm, $h = 10.5 = \frac{21}{2}$ cm. Cup: $r = 3.5$ cm. Tent: $R = 7$ m, used CSA $= 551 - 1 = 550$ m^2.

Formula / Theorem Used: $V_{\text{cyl}} = \pi r^2 h$, $V_{\text{hemi}} = \frac{2}{3}\pi r^3$, $\text{CSA}_{\text{cone}} = \pi R l$

Sub-part (i): Cylindrical Glass Capacity [1 Mark]

$V_{\text{glass}} = \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{21}{2} = \frac{11\times147}{4} = \frac{1617}{4} = \boxed{\textbf{404.25 cm}^3}$.

Sub-part (ii): Hemispherical Cup Capacity [1 Mark]

$V_{\text{cup}} = \frac{2}{3} \times \frac{22}{7} \times \frac{7}{2} \times \frac{7}{2} \times \frac{7}{2} = \frac{11\times49}{6} = \frac{539}{6} = \boxed{\textbf{89}\frac{\textbf{5}}{\textbf{6}}\ \textbf{cm}^3\ (\approx \textbf{89.83 cm}^3)}$.

Sub-part (iii): Height of Conical Tent [1 Mark]

CSA $= 550 \implies \frac{22}{7} \times 7 \times l = 550 \implies 22l = 550 \implies l = 25$ m.

$H = \sqrt{l^2 - R^2} = \sqrt{25^2 - 7^2} = \sqrt{625 - 49} = \sqrt{576} = \boxed{\textbf{24 m}}$.

Sub-part (iv): Floor Area per Student (4 Students) [1 Mark]

Total floor area $= \pi R^2 = \frac{22}{7} \times 7 \times 7 = 154$ m$^2 \implies$ Area per student $= \frac{154}{4} = \boxed{\textbf{38.5 m}^2}$.

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